hypothesis1

February 11, 2026

1 Hypothesis 1: Shift in Tipping Culture

We attempt to answer the question: After adjusting for confounding factors such as trip duration and surcharges, do customers in 2019 exhibit different tipping behavior than those in 2014?

[145]:
```
# Load data
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('data/taxidata_clean.csv',␣
 ↪parse_dates=['trip_start_timestamp'])
df['year'] = df['trip_start_timestamp'].dt.year
df['pct_tip'] = 100 * df['tips'] / df['fare'].replace(0, np.nan)
df['hour'] = df['trip_start_timestamp'].dt.hour
df['dow'] = df['trip_start_timestamp'].dt.dayofweek
df['speed_mph'] = 3600 * df['trip_miles'] / df['trip_seconds'].replace(0, np.
 ↪nan)
```

2 1. Exploratory Data Analysis and additional Preprocessing

As seen below, the dataset seemingly did not include any cash tips, as most tips came from riders who paid using credit card.

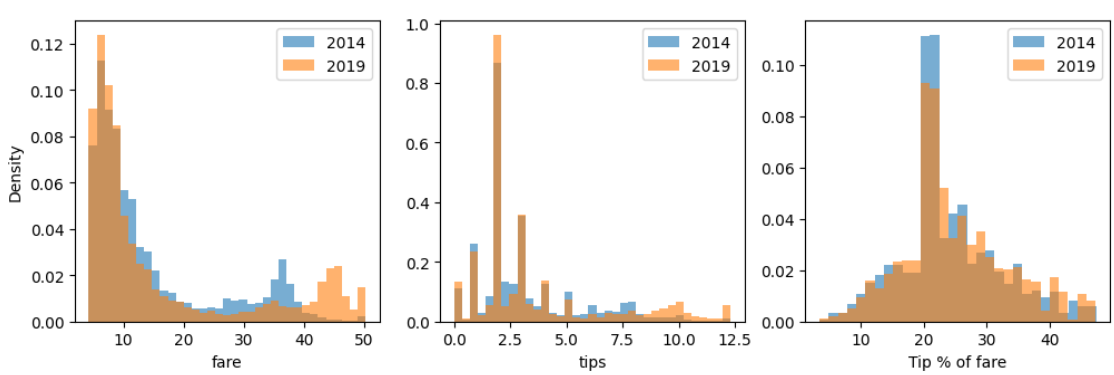
[146]:
```
c = df.assign(has_tip=df['tips']>0).groupby(['year','payment_type']).agg(
    n=('tips','count'), n_tips=('has_tip','sum')
).assign(pct_tipped=lambda x: 100*x['n_tips']/x['n'])
print('2014:'); display(c.loc[2014])
print('2019:'); display(c.loc[2019])
```

```
2014:
                   n  n_tips  pct_tipped
payment_type
Cash          128779       1      0.0008
Credit Card    68575   66225     96.5731
Dispute          150       0      0.0000
No Charge       1483       0      0.0000
Unknown          295     113     38.3051
```

```
2019:

                   n  n_tips  pct_tipped
payment_type
Cash           93679      46      0.0491
Credit Card    98698   94319     95.5632
Dispute           69       0      0.0000
Mobile          3480    3433     98.6494
No Charge        319       5      1.5674
Pcard              2       0      0.0000
Prcard          2085     158      7.5779
Prepaid            9       0      0.0000
Unknown         1314      39      2.9680
```

[147]:
```
# CC users only: fare, tip, and tip % distributions, 2014 vs 2019
cc = df[(df['payment_type']=='Credit Card') & (df['fare']>0)].copy()
cc_pct = cc[cc['tips']>0][['year','pct_tip']].dropna()
q1, q3 = cc_pct['pct_tip'].quantile([0.25, 0.75])
cc_pct = cc_pct[(cc_pct['pct_tip']>=q1-1.5*(q3-q1)) & (cc_pct['pct_tip']<=q3+1.
↪5*(q3-q1))]

fig, ax = plt.subplots(1, 3, figsize=(12, 3.5))
for col, ax_ in zip(['fare', 'tips'], ax[:2]):
    for y in [2014, 2019]:
        s = cc[cc['year']==y][col].clip(cc[col].quantile(0.01), cc[col].
↪quantile(0.99))
        ax_.hist(s, bins=35, alpha=0.6, label=y, density=True)
    ax_.set_xlabel(col); ax_.legend()
for y in [2014, 2019]:
    ax[2].hist(cc_pct[cc_pct['year']==y]['pct_tip'], bins=30, alpha=0.6,␣
↪label=y, density=True)
ax[2].set_xlabel('Tip % of fare'); ax[2].legend()
ax[0].set_ylabel('Density')
```

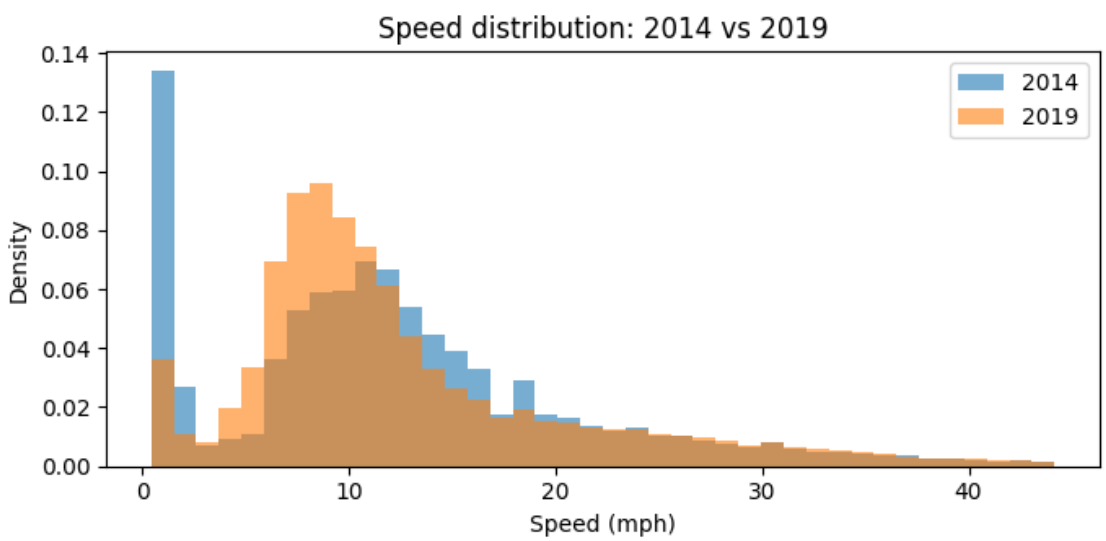
[147]: Text(0, 0.5, 'Density')



2.0.1 Speed: 2014 vs 2019

Speed (mph) = trip_miles / (trip_seconds/3600). Trips with trip_seconds 0 excluded.

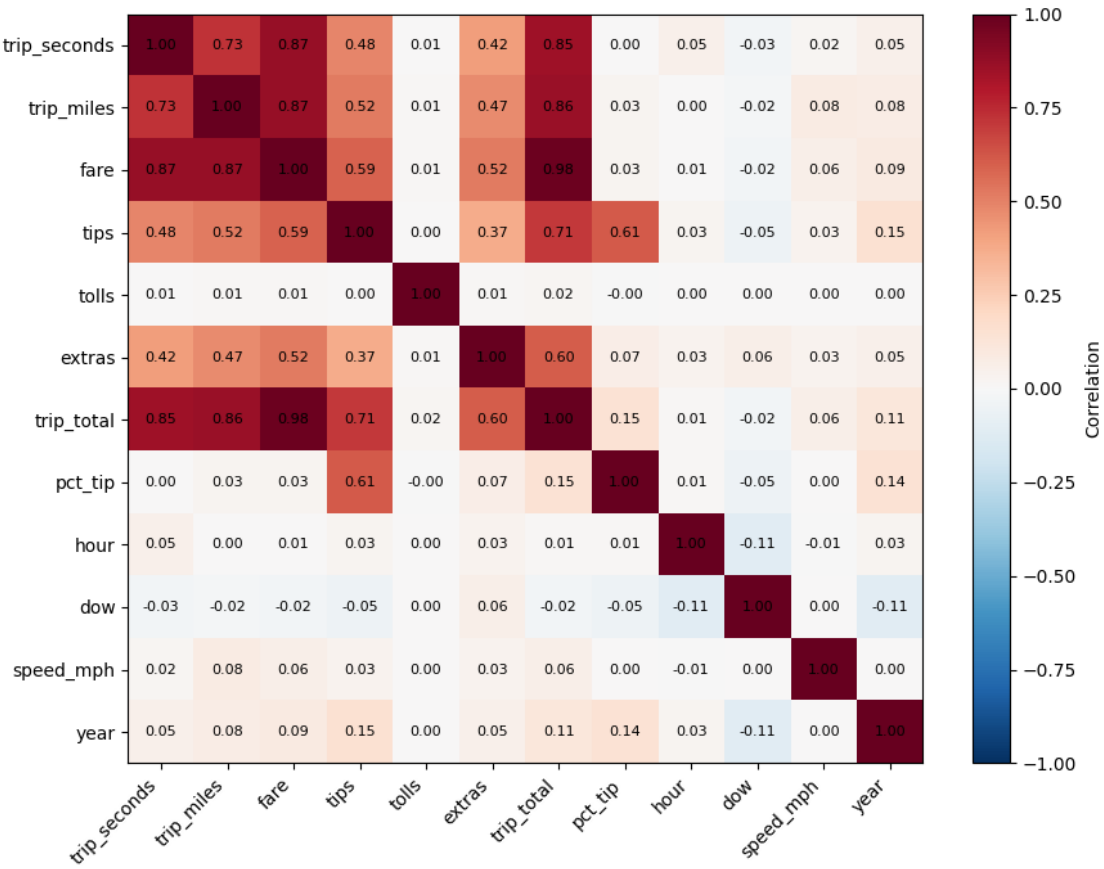
[148]:
```
sp = df[df['trip_seconds']>0][['year','speed_mph']].dropna()
sp = sp[sp['speed_mph'].between(sp['speed_mph'].quantile(0.01), sp['speed_mph'].
 ↪quantile(0.99))]

fig, ax = plt.subplots(figsize=(7, 3.5))
for y in [2014, 2019]:
    ax.hist(sp[sp['year']==y]['speed_mph'], bins=40, alpha=0.6, label=y,␣
 ↪density=True)
ax.set_xlabel('Speed (mph)'); ax.set_ylabel('Density'); ax.legend()
ax.set_title('Speed distribution: 2014 vs 2019')
plt.tight_layout(); plt.show()

y14, y19 = sp[sp['year']==2014]['speed_mph'], sp[sp['year']==2019]['speed_mph']
print(f'2014: mean={y14.mean():.2f} mph, median={y14.median():.2f}')
print(f'2019: mean={y19.mean():.2f} mph, median={y19.median():.2f}')
```




```
2014: mean=12.35 mph, median=11.22
2019: mean=12.74 mph, median=10.34
```

[149]:
```
num_cols = ['trip_seconds', 'trip_miles', 'fare', 'tips', 'tolls', 'extras',␣
 ↪'trip_total', 'pct_tip', 'hour', 'dow', 'speed_mph', 'year']
corr = df[num_cols].corr()
```

```
fig, ax = plt.subplots(figsize=(9, 7))
im = ax.imshow(corr, cmap='RdBu_r', vmin=-1, vmax=1, aspect='auto')
ax.set_xticks(range(len(num_cols)))
ax.set_yticks(range(len(num_cols)))
ax.set_xticklabels(num_cols, rotation=45, ha='right')
ax.set_yticklabels(num_cols)
for i in range(len(num_cols)):
    for j in range(len(num_cols)):
        ax.text(j, i, f'{corr.iloc[i,j]:.2f}', ha='center', va='center',␣
 ↪fontsize=8)
plt.colorbar(im, ax=ax, label='Correlation')
plt.tight_layout()
plt.show()
```



To choose confouding factors, we take a look at the available variables and decide on the `trip_seconds`, `trip_miles`, `fare`, `extras` and `speed_mph`.

2.0.2 Preprocessing validation: confounding variables

Check that `trip_seconds`, `trip_miles`, `fare`, `extras`, `speed_mph` are valid for the analysis subset (CC, fare>0). Drop invalid rows. Use `df_cc` for subsequent modeling.

[150]:
```
confound_cols = ['trip_seconds', 'trip_miles', 'fare', 'extras', 'speed_mph']
mask_cc = (df['payment_type'] == 'Credit Card') & (df['fare'] > 0)
df_cc = df[mask_cc].copy()

print('Validity checks (CC, fare>0):')
for c in confound_cols:
    n_na = df_cc[c].isna().sum()
    n_neg = (df_cc[c] < 0).sum()
    n_zero = (df_cc[c] == 0).sum() if c != 'extras' else 'N/A'
    print(f'  {c:14s}: NaN={n_na:,}, neg={n_neg:,}', end='')
    if c != 'extras':
        print(f', zero={n_zero:,}')
    else:
        print()

# Additional filters: drop rows with invalid confounding vars
valid = (
    (df_cc['trip_seconds'] > 0) &
    (df_cc['trip_miles'] > 0) &
    (df_cc['fare'] > 0) &
    (df_cc['extras'] >= 0) &
    (df_cc['speed_mph'].notna()) &
    (df_cc['speed_mph'] > 0) &
    (df_cc['speed_mph'] <= 100)
)
dropped = df_cc[~valid]
df_cc = df_cc[valid]
n_drop = len(dropped)
print(f'\nDropped {n_drop} rows with invalid confounders. Final n={len(df_cc):
 ↪,}')
```

```
Validity checks (CC, fare>0):
  trip_seconds  : NaN=0, neg=0, zero=0
  trip_miles    : NaN=0, neg=0, zero=0
  fare          : NaN=0, neg=0, zero=0
  extras        : NaN=0, neg=0
  speed_mph     : NaN=0, neg=0, zero=0

Dropped 32 rows with invalid confounders. Final n=167,241
```

3 2. Methodology

3.1 Global Test — Doubly Robust Local Permutation

We test the global null hypothesis:

$$H_0 : \text{Year} \perp \text{tips} \mid Z = (\text{fare, extras, trip_seconds, trip_miles, speed_mph})$$

Steps: 1. Fit OLS: `tips ~ fare + extras + trip_seconds + trip_miles + speed_mph`. Compute residuals $\hat{r}_i$. 2. Test statistic: $T = \bar{\hat{r}}_{2019} - \bar{\hat{r}}_{2014}$ 3. Create bins by 10% quantiles of fare (10 bins). Permute year labels within each bin to generate null distribution. 4. Two-sided p-value with $+1/+1$ correction.

3.1.1 Step 1: Outcome regression (no year)

[ ]:
```
import statsmodels.api as sm
from scipy import stats

SEED = 42
M_PERM = 1000
ALPHA = 0.05
rng = np.random.default_rng(SEED)

df_cc['year_dummy'] = (df_cc['year'] == 2019).astype(int)
CONFOUNDERS = ['fare', 'extras', 'trip_seconds', 'trip_miles', 'speed_mph']

#   Step 1: OLS
X_no_year = df_cc[CONFOUNDERS].copy()
X_no_year.insert(0, 'const', 1.0)
Y = df_cc['tips'].values

ols_no_year = sm.OLS(Y, X_no_year).fit()
df_cc['residual'] = ols_no_year.resid

#   Bins: 10% quantiles of fare
fare_deciles = pd.qcut(df_cc['fare'], q=10, labels=False, duplicates='drop')
bin_labels = np.array([f'D{int(x):02d}' for x in fare_deciles + 1])
df_cc['perm_bin'] = pd.Categorical(bin_labels, categories=sorted(np.
 ↪unique(bin_labels)), ordered=True)

print(ols_no_year.summary())
print(f"\nNumber of bins: {df_cc['perm_bin'].nunique()}")
print(f"\nObservations per bin:")
bin_counts = df_cc.groupby('perm_bin', observed=True).agg(
    n=('year_dummy', 'count'),
    n_2014=('year_dummy', lambda x: (x == 0).sum()),
    n_2019=('year_dummy', 'sum')
).assign(p_2019=lambda x: x['n_2019'] / x['n']).sort_index(ascending=True)
```

```python
print(bin_counts.to_string())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.683
Model:                            OLS   Adj. R-squared:                  0.683
Method:                 Least Squares   F-statistic:                 7.194e+04
Date:                Wed, 11 Feb 2026   Prob (F-statistic):               0.00
Time:                        20:28:41   Log-Likelihood:            -3.1837e+05
No. Observations:              167241   AIC:                         6.368e+05
Df Residuals:                  167235   BIC:                         6.368e+05
Df Model:                           5
Covariance Type:            nonrobust
===================================================================================
                    coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const             0.8246      0.012     66.259      0.000       0.800       0.849
fare              0.1700      0.001    182.117      0.000       0.168       0.172
extras            0.1576      0.003     53.308      0.000       0.152       0.163
trip_seconds -9.678e-05   1.34e-05     -7.216      0.000      -0.000   -7.05e-05
trip_miles        0.0226      0.002      9.703      0.000       0.018       0.027
speed_mph        -0.0107      0.001    -11.964      0.000      -0.012      -0.009
==============================================================================
Omnibus:                   202938.265   Durbin-Watson:                   1.998
Prob(Omnibus):                  0.000   Jarque-Bera (JB):       1276012244.174
Skew:                           5.237   Prob(JB):                         0.00
Kurtosis:                     430.791   Cond. No.                     4.08e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 4.08e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

Number of bins: 10

Observations per bin:
              n  n_2014  n_2019  p_2019
perm_bin
D01       18803    6852   11951  0.6356
D02       15350    5758    9592  0.6249
D03       19453    7353   12100  0.6220
D04       16171    6450    9721  0.6011
D05       15109    6315    8794  0.5820
D06       15839    8008    7831  0.4944
D07       16381    7787    8594  0.5246
D08       16704    8016    8688  0.5201
```

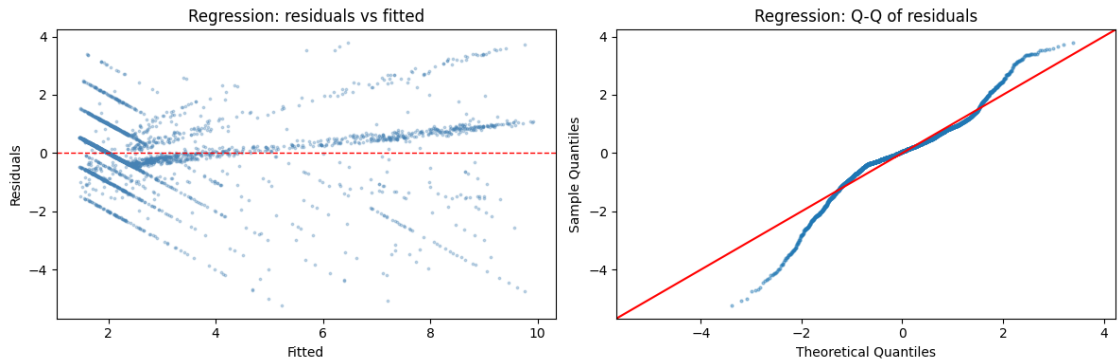
```
D09        16716   10315    6401  0.3829
D10        16715    1705   15010  0.8980
```

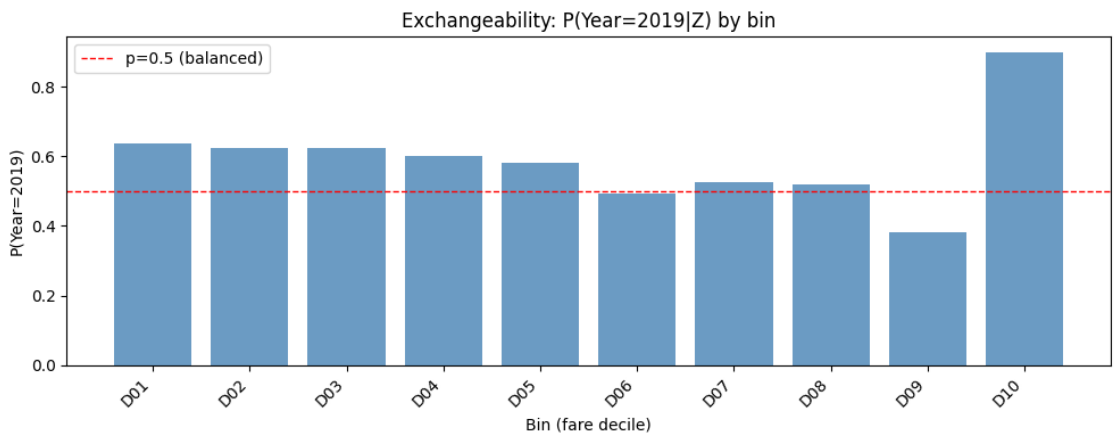
3.1.2 Visual assumption checks: regression & exchangeability

[152]:
```
#   Brief visual checks: regression & exchangeability

# Subsample and limit outliers for clearer visualization
resid, fitted = ols_no_year.resid.values, ols_no_year.fittedvalues.values
idx = np.random.RandomState(SEED).choice(len(resid), size=min(3000,␣
 ↪len(resid)), replace=False)
r, f = resid[idx], fitted[idx]
# Clip to 1st-99th percentile to reduce outlier dominance
r_lo, r_hi = np.percentile(r, 1), np.percentile(r, 99)
f_lo, f_hi = np.percentile(f, 1), np.percentile(f, 99)
mask = (r >= r_lo) & (r <= r_hi) & (f >= f_lo) & (f <= f_hi)
r_plot, f_plot = r[mask], f[mask]

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# 1. Regression: residuals vs fitted & Q-Q
axes[0].scatter(f_plot, r_plot, s=3, alpha=0.3, color='steelblue')
axes[0].axhline(0, color='red', ls='--', lw=1)
axes[0].set_xlabel('Fitted'); axes[0].set_ylabel('Residuals')
axes[0].set_title('Regression: residuals vs fitted')

# Q-Q on subsample
sm.qqplot(r_plot, line='45', ax=axes[1], markersize=2, alpha=0.4,␣
 ↪color='steelblue')
axes[1].set_title('Regression: Q-Q of residuals')

plt.tight_layout(); plt.show()

fig, ax = plt.subplots(figsize=(10, 4))
bc = bin_counts.reset_index().sort_values('perm_bin', ascending=True)
bars = ax.bar(range(len(bc)), bc['p_2019'], color='steelblue', alpha=0.8)
ax.axhline(0.5, color='red', ls='--', lw=1, label='p=0.5 (balanced)')
ax.set_xticks(range(len(bc)))
ax.set_xticklabels(bc['perm_bin'], rotation=45, ha='right')
ax.set_ylabel('P(Year=2019)'); ax.set_xlabel('Bin (fare decile)')
ax.set_title('Exchangeability: P(Year=2019|Z) by bin')
ax.legend()
plt.tight_layout(); plt.show()
```

```
/opt/miniconda3/envs/FinM/lib/python3.13/site-
packages/statsmodels/graphics/gofplots.py:1041: UserWarning: color is
redundantly defined by the 'color' keyword argument and the fmt string "b" (->
color=(0.0, 0.0, 1.0, 1)). The keyword argument will take precedence.
```

```
ax.plot(x, y, fmt, **plot_style)
```





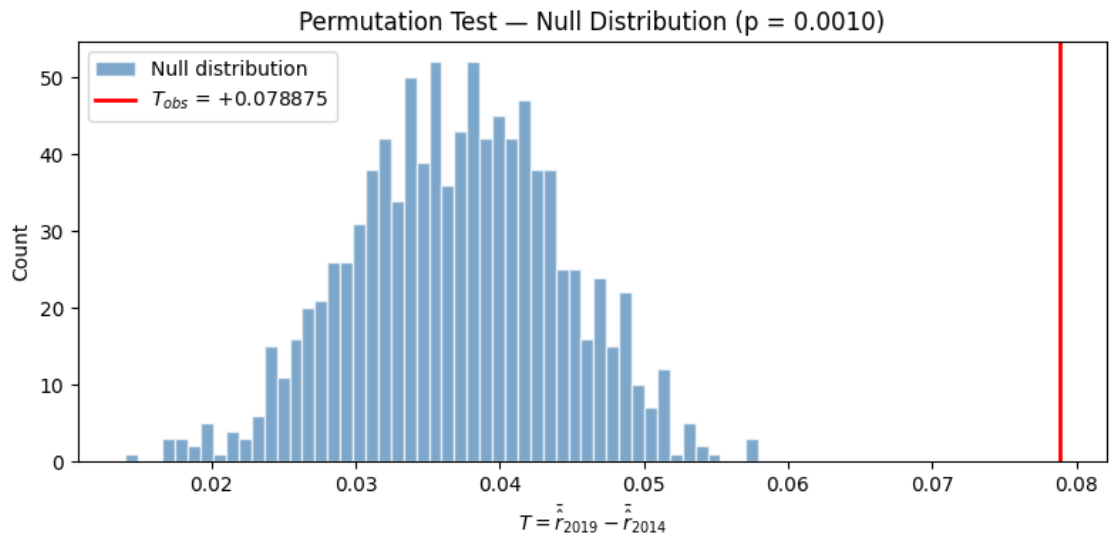
3.1.3 Steps 2–4: Local permutation test

[ ]:
```
#  Local permutation test
residuals = df_cc['residual'].values
year_labels = df_cc['year_dummy'].values
bin_codes, _ = pd.factorize(df_cc['perm_bin'])

# Pre-compute indices per bin (only bins with both years)
bin_indices = {}
for b in np.unique(bin_codes):
    idx = np.where(bin_codes == b)[0]
    if len(np.unique(year_labels[idx])) == 2:
        bin_indices[b] = idx

# Observed test statistic  T = mean | 2019 - mean | 2014
T_obs = residuals[year_labels == 1].mean() - residuals[year_labels == 0].mean()
```

```
# Permute year labels within bins
null_T = np.empty(M_PERM)
for m in range(M_PERM):
    shuffled_years = year_labels.copy()
    for b, idx in bin_indices.items():
        block = shuffled_years[idx]
        rng.shuffle(block)
        shuffled_years[idx] = block
    null_T[m] = residuals[shuffled_years == 1].mean() -␣
 ↪residuals[shuffled_years == 0].mean()

# Two-sided p-value with +1/+1 correction
p_global = (1 + np.sum(np.abs(null_T) >= np.abs(T_obs))) / (1 + M_PERM)

print(f"T_obs (mean resid 2019 - mean resid 2014): {T_obs:+.6f}")
print(f"p-value (two-sided):                       {p_global:.4f}")
if p_global < ALPHA:
    print(f"=> REJECT H0 at alpha={ALPHA}")
else:
    print(f"=> Fail to reject H0 at alpha={ALPHA}")
```

```
T_obs (mean resid 2019 - mean resid 2014): +0.078875
p-value (two-sided):                       0.0010
=> REJECT H0 at alpha=0.05
```

[ ]:
```
fig, ax = plt.subplots(figsize=(8, 4))
ax.hist(null_T, bins=50, edgecolor='white', alpha=0.7, color='steelblue',␣
 ↪label='Null distribution')
ax.axvline(T_obs, color='red', lw=2, label=f'$T_{{obs}}$ = {T_obs:+.6f}')
ax.set_xlabel('$T = \\bar{\\hat{r}}_{2019} - \\bar{\\hat{r}}_{2014}$')
ax.set_ylabel('Count')
ax.set_title(f'Permutation Test - Null Distribution (p = {p_global:.4f})')
ax.legend()
plt.tight_layout()
plt.show()
```



3.2 Stage B: Per-Community-Area Tests

A natural follow-up: **in which areas** did tipping behavior change?

For each community area $i = 1, \ldots, m$: 1. Subset to rides from area i 2. Fit a local regression (same covariates, no year), compute residuals 3. Run the same doubly-robust local permutation test (bins = 10% quantiles of fare, same as before) 4. Obtain p-value p_j

[155]:
```
# Chicago community area names
CA_NAMES = {
    1: "Rogers Park", 2: "West Ridge", 3: "Uptown", 4: "Lincoln Square",
    5: "North Center", 6: "Lake View", 7: "Lincoln Park", 8: "Near North Side",
    9: "Edison Park", 10: "Norwood Park", 11: "Jefferson Park",
    12: "Forest Glen", 13: "North Park", 14: "Albany Park",
    15: "Portage Park", 16: "Irving Park", 17: "Dunning",
    18: "Montclare", 19: "Belmont Cragin", 20: "Hermosa",
    21: "Avondale", 22: "Logan Square", 23: "Humboldt Park",
    24: "West Town", 25: "Austin", 26: "West Garfield Park",
    27: "East Garfield Park", 28: "Near West Side", 29: "North Lawndale",
    30: "South Lawndale", 31: "Lower West Side", 32: "Loop",
    33: "Near South Side", 34: "Armour Square", 35: "Douglas",
    36: "Oakland", 37: "Fuller Park", 38: "Grand Boulevard",
    39: "Kenwood", 40: "Washington Park", 41: "Hyde Park",
    42: "Woodlawn", 43: "South Shore", 44: "Chatham",
    45: "Avalon Park", 46: "South Chicago", 47: "Burnside",
    48: "Calumet Heights", 49: "Roseland", 50: "Pullman",
    51: "South Deering", 52: "East Side", 53: "West Pullman",
    54: "Riverdale", 55: "Hegewisch", 56: "Garfield Ridge",
    57: "Archer Heights", 58: "Brighton Park", 59: "McKinley Park",
```

```
    60: "Bridgeport", 61: "New City", 62: "West Elsdon",
    63: "Gage Park", 64: "Clearing", 65: "West Lawn",
    66: "Chicago Lawn", 67: "West Englewood", 68: "Englewood",
    69: "Greater Grand Crossing", 70: "Ashburn", 71: "Auburn Gresham",
    72: "Beverly", 73: "Washington Heights", 74: "Mount Greenwood",
    75: "Morgan Park", 76: "O'Hare", 77: "Edgewater",
}

MIN_PER_YEAR = 50

# Drop rows with missing pickup area
df_cc = df_cc.dropna(subset=['pickup_community_area'])
df_cc['pickup_community_area'] = df_cc['pickup_community_area'].astype(int)

# Check which areas qualify
area_counts = df_cc.groupby(['pickup_community_area', 'year']).size().
 ↪unstack(fill_value=0)
qualified = area_counts[(area_counts[2014] >= MIN_PER_YEAR) &␣
 ↪(area_counts[2019] >= MIN_PER_YEAR)]
print(f"Areas with >= {MIN_PER_YEAR} rides in both years: {len(qualified)} /␣
 ↪{df_cc['pickup_community_area'].nunique()}")
print(f"Total rides in qualified areas: {df_cc[df_cc['pickup_community_area'].
 ↪isin(qualified.index)].shape[0]:,}")
```

```
Areas with >= 50 rides in both years: 21 / 70
Total rides in qualified areas: 166,089
```

[ ]:
```
def local_perm_test(data, confounders=CONFOUNDERS, m_perm=M_PERM, rng=rng):
    """
    Doubly-robust local permutation test for a single area.

    Returns dict with T_obs, p_value, sample sizes, raw effect, null␣
 ↪distribution.
    """
    Y_loc = data['tips'].values
    T_loc = data['year_dummy'].values

    # Local regression (same covariates, no year)
    X_loc = data[confounders].copy().reset_index(drop=True)
    X_loc.insert(0, 'const', 1.0)

    try:
        model = sm.OLS(Y_loc, X_loc).fit()
        resid = model.resid
    except Exception:
        return None
```

```
    # Local bins: same as Stage A - 10% quantiles of fare
    try:
        fare_deciles = pd.qcut(data['fare'], q=10, labels=False,␣
↪duplicates='drop')
    except ValueError:
        fare_deciles = pd.qcut(data['fare'], q=2, labels=False,␣
↪duplicates='drop')
    bin_codes_loc = fare_deciles

    # Pre-compute bin indices (bins with both years)
    local_bin_idx = {}
    for b in np.unique(bin_codes_loc):
        idx = np.where(bin_codes_loc == b)[0]
        if len(np.unique(T_loc[idx])) == 2:
            local_bin_idx[b] = idx

    # Observed test statistic
    T_obs_loc = resid[T_loc == 1].mean() - resid[T_loc == 0].mean()

    # Permutation null
    null_vals = np.empty(m_perm)
    for i in range(m_perm):
        sh = T_loc.copy()
        for b, idx in local_bin_idx.items():
            block = sh[idx]
            rng.shuffle(block)
            sh[idx] = block
        null_vals[i] = resid[sh == 1].mean() - resid[sh == 0].mean()

    p_val = (1 + np.sum(np.abs(null_vals) >= np.abs(T_obs_loc))) / (1 + m_perm)

    effect_raw = Y_loc[T_loc == 1].mean() - Y_loc[T_loc == 0].mean()

    return {
        'T_obs': T_obs_loc,
        'p_value': p_val,
        'n_2014': int((T_loc == 0).sum()),
        'n_2019': int((T_loc == 1).sum()),
        'effect_raw': effect_raw,
        'null_dist': null_vals,
    }

# Run for all qualified areas
results = []
for area in qualified.index:
    subset = df_cc[df_cc['pickup_community_area'] == area].copy().
↪reset_index(drop=True)
```

```
    out = local_perm_test(subset)
    if out is None:
        continue
    out['area_id'] = area
    out['area_name'] = CA_NAMES.get(area, f'Area {area}')
    results.append(out)

print(f"Successfully tested {len(results)} areas")
```

```
Successfully tested 21 areas
```

3.3 Stage C: Multiple Testing Corrections

3.3.1 Bonferroni correction (naive)

Reject H_j if $p_j \leq \alpha/m$. This controls the **family-wise error rate** (FWER) $\leq \alpha$, but is conservative — it may miss real effects.

3.3.2 Benjamini-Hochberg (BH) procedure

Sort p-values: $p_{(1)} \leq \cdots \leq p_{(m)}$. Reject $p_{(1)}, \ldots, p_{(\hat{k})}$ where $\hat{k} = \max\{k : p_{(k)} \leq \frac{\alpha k}{m}\}$.

[171]:
```
#   Build results dataframe

res_df = pd.DataFrame([{
    'area_id': r['area_id'],
    'area_name': r['area_name'],
    'n_2014': r['n_2014'],
    'n_2019': r['n_2019'],
    'n': r['n_2014'] + r['n_2019'],
    'T_obs': r['T_obs'],
    'effect_raw': r['effect_raw'],
    'raw_p': r['p_value'],
} for r in results])

res_df = res_df.sort_values('raw_p').reset_index(drop=True)
m = len(res_df)
res_df['rank'] = np.arange(1, m + 1)

#   1. Bonferroni correction
res_df['bonf_adj_p'] = np.minimum(res_df['raw_p'] * m, 1.0)
res_df['bonf_reject'] = res_df['raw_p'] <= ALPHA / m
n_bonf = res_df['bonf_reject'].sum()

print(f"Number of hypotheses (m): {m}")
print(f"\n--- Bonferroni (FWER control) ---")
print(f"Threshold: alpha/m = {ALPHA}/{m} = {ALPHA/m:.4f}")
print(f"Bonferroni rejections: {n_bonf}")
```

```
#   2. BH procedure
res_df['bh_threshold'] = res_df['rank'] / m * ALPHA

# BH: reject all p_(k) where p_(k) <= alpha * k / m, up to max such k
res_df['bh_reject'] = res_df['raw_p'] <= res_df['bh_threshold']
if res_df['bh_reject'].any():
    k_hat = res_df[res_df['bh_reject']].index.max() + 1  # +1 because 0-indexed
    res_df.loc[:res_df[res_df['bh_reject']].index.max(), 'bh_reject'] = True
else:
    k_hat = 0

# BH-adjusted p-values
res_df['bh_adj_p'] = np.minimum(res_df['raw_p'] * m / res_df['rank'], 1.0)
res_df['bh_adj_p'] = res_df['bh_adj_p'][::-1].cummin()[::-1]

print(f"\n--- Benjamini-Hochberg (FDR control) ---")
print(f"BH discoveries (k_hat): {k_hat}")
print(f"FDR control level:      {ALPHA}")
```

```
Number of hypotheses (m): 21

--- Bonferroni (FWER control) ---
Threshold: alpha/m = 0.05/21 = 0.0024
Bonferroni rejections: 3

--- Benjamini-Hochberg (FDR control) ---
BH discoveries (k_hat): 3
FDR control level:      0.05
```

4 3. Results

[165]:
```
#   Results table
display_cols = ['area_name', 'n', 'effect_raw', 'T_obs', 'raw_p',
                'bonf_adj_p', 'bonf_reject', 'bh_adj_p', 'bh_reject']

pd.set_option('display.float_format', lambda x: f'{x:.4f}')
res_df[display_cols]
```

[165]:

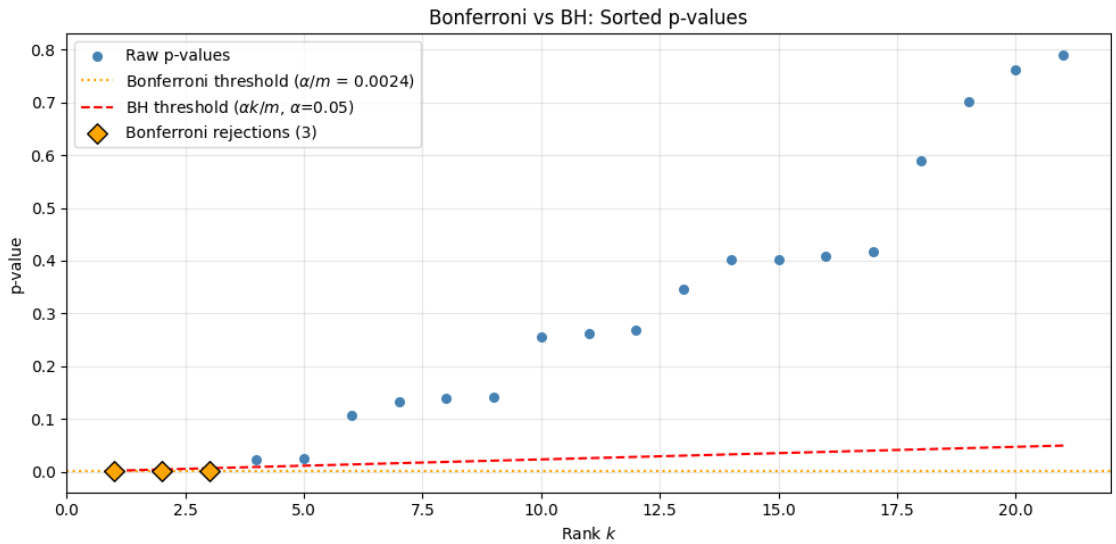
	area_name	n	effect_raw	T_obs	raw_p	bh_adj_p	bh_reject
0	Near North Side	51765	0.0111	0.0666	0.0010	0.0105	True
1	O'Hare	18564	1.5197	0.2225	0.0010	0.0105	True
2	Near West Side	16324	-0.0773	0.0652	0.0020	0.0140	True
3	Lake View	7983	0.4290	0.0710	0.0230	0.1091	False
4	Garfield Ridge	5134	1.2749	0.2002	0.0260	0.1091	False
5	North Center	483	1.1404	0.2603	0.1069	0.3287	False
6	Hyde Park	686	0.8331	0.2174	0.1329	0.3287	False
7	Uptown	1579	0.4451	0.1087	0.1399	0.3287	False

```
8     Armour Square    141      1.0332  0.3324 0.1409    0.3287      False
9   Near South Side   5191      0.4945  0.0565 0.2547    0.4685      False
10        West Town   3275      0.3524  0.0899 0.2627    0.4685      False
11      Irving Park    265      0.9957 -0.2570 0.2677    0.4685      False
12       West Ridge    182      0.4607  0.4117 0.3457    0.5146      False
13      Albany Park    147      0.6997  0.2257 0.4016    0.5146      False
14     Logan Square   1002      0.3725 -0.1310 0.4026    0.5146      False
15        Edgewater    992      0.4077  0.0997 0.4076    0.5146      False
16   Lincoln Square    391      0.2142 -0.1211 0.4166    0.5146      False
17             Loop  45400     -0.1456  0.0087 0.5894    0.6876      False
18     Lincoln Park   6071      0.2862  0.0154 0.7023    0.7762      False
19          Douglas    177      0.4081  0.0955 0.7612    0.7902      False
20      Rogers Park    337      0.0030  0.0599 0.7902    0.7902      False
```

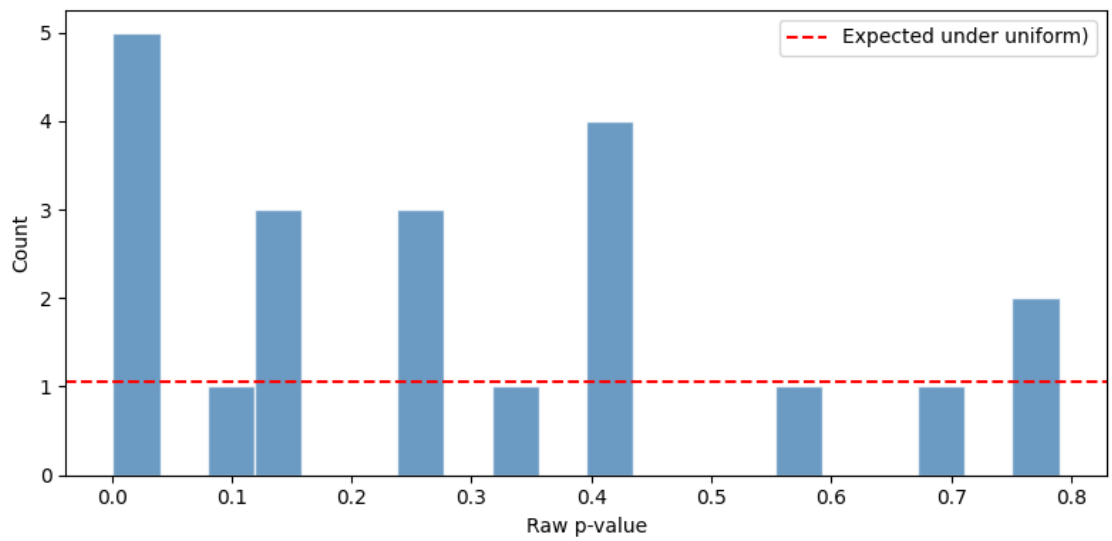
[176]:

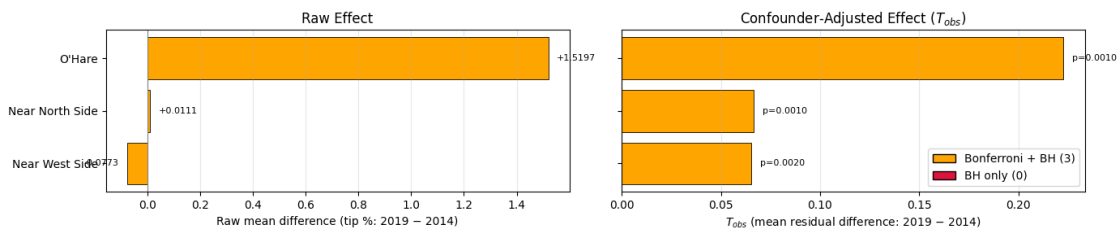
```python
#  sorted p-values vs Bonferroni & BH thresholds
fig, ax = plt.subplots(figsize=(10, 5))
ax.scatter(res_df['rank'], res_df['raw_p'], color='steelblue', s=30, zorder=3,␣
 ↪label='Raw p-values')

# Bonferroni threshold
ax.axhline(ALPHA / m, color='orange', ls=':', lw=1.5,
           label=f'Bonferroni threshold ($\\alpha/m$ = {ALPHA/m:.4f})')

# BH threshold
ax.plot(res_df['rank'], res_df['bh_threshold'], color='red', ls='--', lw=1.5,
        label=f'BH threshold ($\\alpha k / m$, $\\alpha$={ALPHA})')

# Highlight Bonferroni rejections
if n_bonf > 0:
    ax.scatter(res_df[res_df['bonf_reject']]['rank'],
               res_df[res_df['bonf_reject']]['raw_p'],
               color='orange', s=80, zorder=5, edgecolors='black', marker='D',
               label=f'Bonferroni rejections ({n_bonf})')

# Highlight BH discoveries (those not already caught by Bonferroni)
bh_only = res_df['bh_reject'] & ~res_df['bonf_reject']
if bh_only.any():
    ax.scatter(res_df[bh_only]['rank'],
               res_df[bh_only]['raw_p'],
               color='crimson', s=50, zorder=4, edgecolors='black',
               label=f'BH-only discoveries ({bh_only.sum()})')

ax.set_xlabel('Rank $k$')
ax.set_ylabel('p-value')
ax.set_title(f'Bonferroni vs BH: Sorted p-values')
ax.legend()
ax.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```



[177]:
```
fig, ax = plt.subplots(figsize=(8, 4))
ax.hist(res_df['raw_p'], bins=20, edgecolor='white', color='steelblue', alpha=0.
 ↪8)
ax.axhline(m / 20, color='red', ls='--', lw=1.5, label=f'Expected under␣
 ↪uniform)')
ax.set_xlabel('Raw p-value')
ax.set_ylabel('Count')
ax.legend()
plt.tight_layout()
plt.show()
```



[180]:
```
from matplotlib.patches import Patch

#   Horizontal bar plots: raw mean difference & T_obs for BH discoveries
disc = res_df[res_df['bh_reject']].copy()
disc = disc.sort_values('T_obs')  # order by T statistic

colors = ['orange' if b else 'crimson' for b in disc['bonf_reject']]
h = max(3, 0.7 * len(disc))

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, h), sharey=True)

#   Left panel: raw mean difference
ax1.barh(disc['area_name'], disc['effect_raw'], color=colors,␣
 ↪edgecolor='black', linewidth=0.6)
ax1.axvline(0, color='grey', ls='-', lw=0.8)
ax1.set_xlabel('Raw mean difference (tip %: 2019 - 2014)')
ax1.set_title('Raw Effect')
for i, (_, row) in enumerate(disc.iterrows()):
    offset = 0.02 * (disc['effect_raw'].abs().max())
    x_pos = row['effect_raw'] + (offset if row['effect_raw'] >= 0 else -offset)
    ha = 'left' if row['effect_raw'] >= 0 else 'right'
    ax1.text(x_pos, i, f"{row['effect_raw']:+.4f}", va='center', ha=ha,␣
 ↪fontsize=8)
ax1.grid(True, axis='x', alpha=0.3)

#   Right panel: T statistic (confounder-adjusted)
ax2.barh(disc['area_name'], disc['T_obs'], color=colors, edgecolor='black',␣
 ↪linewidth=0.6)
```

```python
ax2.axvline(0, color='grey', ls='-', lw=0.8)
ax2.set_xlabel('$T_{obs}$ (mean residual difference: 2019 - 2014)')
ax2.set_title('Confounder-Adjusted Effect ($T_{obs}$)')
for i, (_, row) in enumerate(disc.iterrows()):
    offset = 0.02 * (disc['T_obs'].abs().max())
    x_pos = row['T_obs'] + (offset if row['T_obs'] >= 0 else -offset)
    ha = 'left' if row['T_obs'] >= 0 else 'right'
    ax2.text(x_pos, i, f"p={row['raw_p']:.4f}", va='center', ha=ha, fontsize=8)
ax2.grid(True, axis='x', alpha=0.3)

# Legend
legend_elements = [
    Patch(facecolor='orange', edgecolor='black', label=f'Bonferroni + BH␣
↪({disc["bonf_reject"].sum()})'),
    Patch(facecolor='crimson', edgecolor='black', label=f'BH only␣
↪({(~disc["bonf_reject"]).sum()})'),
]
ax2.legend(handles=legend_elements, loc='lower right')

plt.tight_layout()
plt.show()
```